Building a Pipeline for Time Series Data

Working with Time Series Data: Formative Assignment

Group 13

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Repository: https://github.com/jbyiringiro/gdp-time-series-pipeline

October 2026

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1. Introduction

This report describes the design and implementation of a pipeline for time-series data built on the Global Economy Indicators dataset (1970–2021). The pipeline consists of four components, which correspond to the four tasks of the assignment:

1. Time-series preprocessing, exploratory data analysis and modelling, which define the prediction problem and produce the trained model (Section 4).
2. A relational database in MySQL and a document database in MongoDB, which store the same records (Section 5).
3. An API that provides create, read, update and delete (CRUD) operations and time-series queries for both databases (Section 6).
4. A prediction script that retrieves records through the API, applies the preprocessing developed in Task 1, loads the trained model and produces a GDP estimate (Section 7).

Sections 2 and 3 define the prediction problem and justify the choice of dataset. Section 8 summarises the results and limitations. Table 1 shows how data flow between the components.

Table 1. Components of the pipeline.

Stage	Component	Input	Output
1	Notebook Building_a_Pipeline_for_Time_Series_Data.ipynb (Task 1)	CSV file	Cleaned series, features, model experiments
2	task2/mysql_setup.py, task2/mongodb_setup.py (Task 2)	CSV file	MySQL tables and MongoDB collection with 10,512 records each
3	task3/api_demo.py (Task 3)	MySQL and MongoDB	JSON responses over HTTP
4	task4/predict_from_api.py (Task 4)	API responses and the saved model	GDP estimate for a selected year

2. Problem Definition

The objective is to estimate the annual Gross Domestic Product (GDP) of the United States, in current US dollars, for a given year t. For each year the model receives the GDP history up to year $t-1$, summarised by the previous year's GDP (GDP_Lag1) and the mean GDP of the five previous years (GDP_5Year_MA), together with two indicators for year t: population and final consumption expenditure. The output is a single value, the GDP of year t. Because two of the inputs refer to the same year as the target, the task is a nowcast, that is, an estimate of the current year's GDP from related indicators, rather than a forecast several years ahead.

Table 2. Specification of the prediction problem.

Item	Definition
Unit of analysis	United States, one observation per year
Target	Gross Domestic Product (GDP), current US$
Type of task	Regression on a time series, one year at a time
Training period	1975–2011 (first 80% of the usable years, in time order)
Test period	2012–2021 (last 20%, including the 2020 COVID-19 shock)
Evaluation metrics	RMSE (US$), MAPE (%), R^2
Success criterion	Lower test error than the naive forecast in which GDP for year t equals GDP for year $t-1$

GDP is the principal measure of the size of an economy. Governments express budgets, debt and deficit targets as a share of GDP, central banks use it to assess whether the economy is expanding or contracting, and firms use it to estimate the size of markets. The official annual figure is published only after the year has ended and is revised subsequently, so a transparent estimate derived from indicators that are already available has practical value for planning. The United States was selected because it is the largest economy in the dataset and has a complete series (Section 3).

The problem is treated as a time-series problem because GDP in one year builds on GDP in the preceding year, so the order of the observations carries information. Three consequences follow: lagged and moving-average features are constructed; the data are divided into training and test sets by time rather than at random; and hyperparameters are tuned on time-ordered folds, so that the model is never trained on observations from the future (Hyndman & Athanasopoulos, 2021).

3. Dataset and Justification

The dataset is Global Economy Indicators, published on Kaggle (prasad22, n.d.) and compiled from the National Accounts Main Aggregates Database of the United Nations Statistics Division (United Nations Statistics Division, n.d.). The database is produced in cooperation with national statistical offices and follows a single accounting standard, the System of National Accounts (European Commission et al., 2009). The file contains 10,512 rows, one per country and year, for 220 countries between 1970 and 2021. Its 26 columns cover GDP, Gross National Income (GNI), population, exchange rates, the expenditure components of GDP (consumption, investment, exports and imports) and value added by economic sector.

Table 3. Assignment requirements and how the dataset meets them.

Requirement	Evidence in the dataset
Clear timestamp or date column	Year (1970–2021), one record per country per year
Meaningful prediction target	Gross Domestic Product (GDP)
Multiple measurable variables over time	22 numeric indicators recorded every year

The dataset was chosen for five reasons.

1. The predictors have a known relationship with the target. GDP measured by expenditure is the sum of consumption, investment and net exports, GDP = C + I + (X − M). The dataset contains each of these components, and for the United States they sum to the reported GDP within 0.53% in every year. The predictive value of the inputs is therefore grounded in national accounting rather than in incidental correlation.
2. All values come from one official source and follow one method, so years and countries are comparable. This is essential for a time series, since a change in the way data are collected would otherwise appear as a change in the economy.
3. The period is long enough to include economic shocks. The 52 years cover several business cycles, including the 2008–09 financial crisis and the 2020 pandemic, which produced the only two annual declines in US GDP. The test period (2012–2021) contains the 2020 shock, so the evaluation is not limited to stable years.
4. The data are nearly complete. Only 0.87% of all values are missing, and 78% of these are in a single column (Changes in inventories) that is not used by the model.
5. The structure suits the entire pipeline. The country-by-year layout maps directly onto the relational schema of Task 2 and onto one MongoDB document per country and year, and annual records can be served by date range through the API of Task 3.

The United States was selected for the following reasons, each of which was verified in the notebook (Figure 1):

- It is the largest economy in the dataset, with a GDP of US$23.3 trillion in 2021 compared with US$17.7 trillion for China, and it accounted for between 21% and 32% of world GDP in every year since 1970 (31.4% in 1970 and 24.3% in 2021).
- Its series is complete: all 52 years are present and none of the modelled columns has a missing value.
- Its GDP is free of currency distortion. Values are recorded in US dollars and the US exchange rate equals 1.0 in every year, whereas for other countries the dollar value of GDP combines real economic change with exchange-rate movements.
- The expenditure identity holds closely: consumption, investment, exports and imports reproduce the reported GDP within 0.53% in every year, and consumption alone averages 80.2% of GDP.

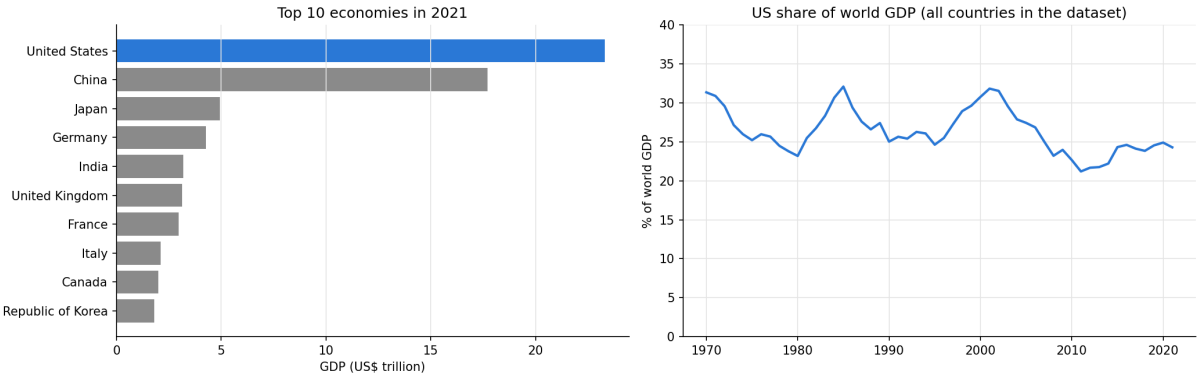


Figure 1. Left: the ten largest economies in 2021, with the United States highlighted. Right: the US share of world GDP in the dataset, 1970–2021. The share moves with the value of the dollar, because the GDP of other countries is converted into dollars.

The dataset also has limitations, which are summarised in Table 4 together with the measures taken to address them.

Table 4. Limitations of the dataset.

Limitation	**Effect**	**Mitigation**
Annual data: 52 observations for one country (47 after feature construction)	Little data for learning; complex models overfit	Few features, simple models, small tuning grids and a strict time-based split
Values in current (nominal) dollars	Growth includes inflation as well as higher output	Nominal GDP is predicted and growth rates are reported as nominal
US values rounded to three significant figures	Changes below about 0.5–1% are partly due to rounding	Differences in error below about 1% are not treated as meaningful

4. Time-Series Preprocessing, Exploratory Analysis and Modelling

This section describes Task 1. All steps are implemented in the notebook Building_a_Pipeline_for_Time_Series_Data.ipynb, which also saves every figure in this section to report/figures/.

4.1 Loading and Cleaning

The CSV file was loaded and the surrounding spaces were removed from the column names and from the text columns Country and Currency. In the raw file every header and text value is padded with spaces (for example, ' Afghanistan '), so without this step a filter on the country name returns no rows and column look-ups by name fail. After cleaning, the table has 26 columns: two identifiers (CountryID and Country), the time column Year, the text column Currency and 22 numeric indicators.

4.2 Time Range and Frequency

The first and last years, the interval between consecutive records of each country, duplicate country-year rows and the number of countries reporting in each year were examined (Figure 2). These checks are necessary because lag and moving-average features assume that each row represents exactly one year; if years were missing or duplicated, a shift of one row would not correspond to the previous year.

The data cover the period 1970 to 2021 (52 years) at annual frequency. Every interval between consecutive records is exactly one year and there are no duplicate country-year rows, so no resampling is required. Of the 220 countries, 179 have all 52 years. The number of reporting countries rises from 187 to 216 in 1990, when the successor states of the USSR, Yugoslavia and Czechoslovakia first appear, and falls to 211 in 1991, when the former states cease to report. In 1990 both the former states and their successors are present, so world totals for that year count these economies twice. This does not affect the US series used for modelling.

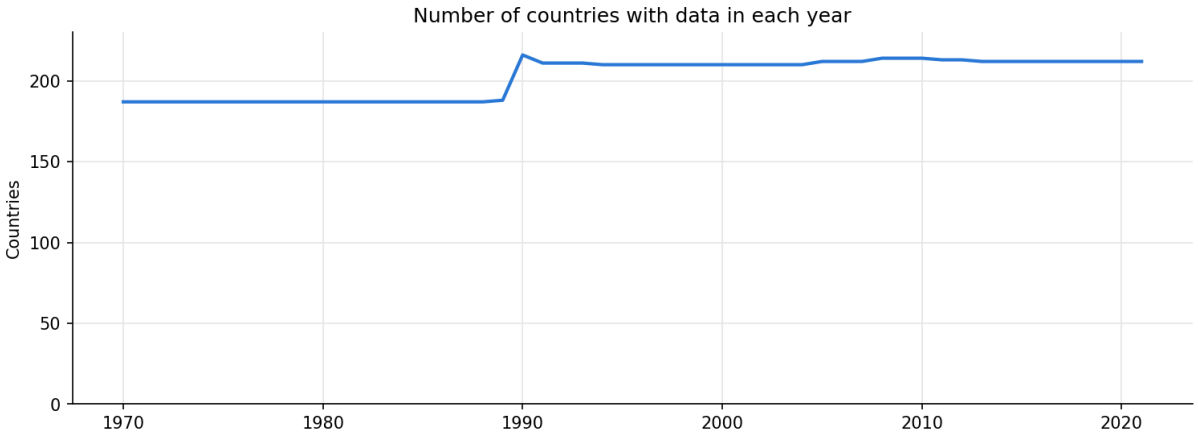


Figure 2. Number of countries with data in each year. The step in 1990 reflects the dissolution of the USSR, Yugoslavia and Czechoslovakia.

4.3 Missing Values

In total, 2,374 values (0.87% of all cells) are missing, and 78% of them are in Changes in inventories, where 17.5% of the rows are empty. Every other column is at least 98.8% complete (Figure 3). The US series contains only two missing values, both in Changes in inventories, for 1970 and 1971.

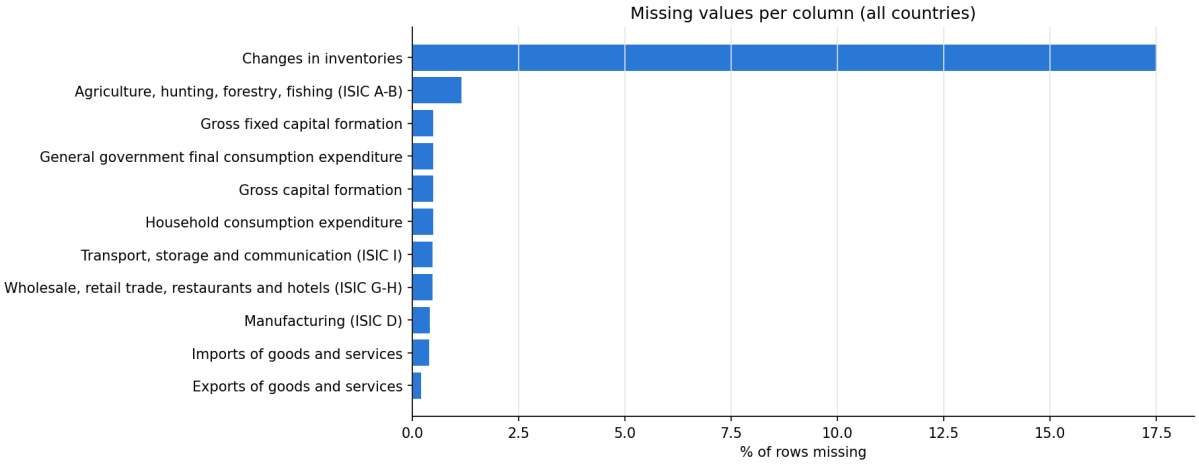


Figure 3. Percentage of missing values per column across all countries. Only columns with at least one missing value are shown.

Missing values were treated as follows:

1. Gaps between two known years were filled by linear interpolation. Economic indicators change gradually from year to year, so a straight line between neighbouring years preserves the trend. Mean imputation would be inappropriate for a trending series: the mean US GDP (US$9.3 trillion) is almost nine times the 1970 value, so a mean-filled value in the 1970s would create a large artificial spike.
2. Gaps at the beginning or end of a series were filled from the nearest known year (backward or forward fill), because interpolation requires a known value on both sides. The two US gaps (1970 and 1971) are of this kind and take the 1972 value.
3. Rows were not removed. Deleting 1970 and 1971 would also discard GDP, population and consumption for those years, although these columns are complete, and would interrupt the continuous sequence of years on which the lag and moving-average features depend.

After this step the US series has no missing values. Since Changes in inventories is not a model input, the filled values do not affect the model.

4.4 Distribution of Numerical Variables

Summary statistics and skewness were computed for the target and the main explanatory variables (Table 5), and their distributions were plotted (Figure 4).

Table 5. Summary statistics for the United States, 1970–2021.

Statistic	**GDP (US$ trillion)**	**Population (million)**	**Consumption (US$ trillion)**	**GNI per capita (US$)**
Mean	9.32	268.1	7.60	32,119
Median	7.86	267.3	6.26	29,213
Standard deviation	6.50	43.3	5.42	18,944
Minimum (1970)	1.07	200.3	0.84	5,363

Statistic	GDP (US$ trillion)	Population (million)	Consumption (US$ trillion)	GNI per capita (US$)
Maximum (2021)	23.30	337.0	19.30	70,081
Skewness	0.48	0.05	0.48	0.28

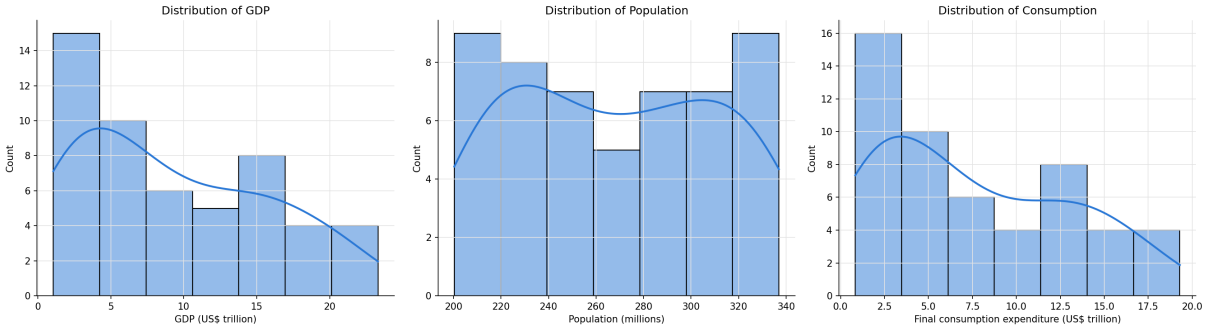


Figure 4. Distributions of GDP, population and final consumption expenditure for the United States, 1970–2021.

GDP is right-skewed (skewness 0.48), with the mean (US$9.32 trillion) above the median (US$7.86 trillion). Consumption has the same shape because it represents a stable share of about 80% of GDP. Population is almost symmetric (skewness 0.05) because it grows by a similar number of people each year, and GNI per capita shows a mild right skew (0.28). For a time series, these distributions describe when values occurred rather than random variation: because GDP grows by a percentage each year, it spends many years at low values and few at high values, which produces the right skew. The mean and variance of GDP therefore depend on the period considered, a property that was tested formally in the next step.

4.5 Stationarity

The Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979), as implemented in statsmodels (Seabold & Perktold, 2010), was applied to the GDP level and to its annual growth rate. The null hypothesis of the test is that the series is non-stationary, that is, that its mean and variance change over time; a p-value below 0.05 rejects this hypothesis.

Table 6. Results of the Augmented Dickey-Fuller test.

Series	ADF p-value	Result
GDP level	0.997	Non-stationary
GDP annual growth rate (%)	0.211	Non-stationary

The GDP level has a strong trend. The growth rate also fails the test, because nominal growth was considerably higher during the high-inflation 1970s (10.5% per year in 1971–1981) than after 2000 (about 4%). These results shaped three later decisions. First, the test years contain GDP values above any value in the training years, so the model must be able to extrapolate, which favours linear models over tree-based models. Second, two trending series always appear correlated, so correlations are examined on growth rates as well as on levels (Questions 3 and 4). Third, the features describe the recent history of the series, so that the model works relative to the current level of GDP.

4.6 Feature Engineering

Table 7 lists the features constructed from the GDP series.

Table 7. Engineered time-series features.

Feature	Definition	Purpose
GDP_Lag1	GDP of year $t-1$ (shift(1))	The strongest single predictor (Question 3)
GDP_5Year_MA	Mean GDP of years $t-5$ to $t-1$ (shift(1) followed by rolling(5))	A stable summary of recent history that smooths single-year shocks (Question 2)
GDP_Growth	Year-on-year change in per cent (pct_change())	Removes the trend; used throughout the analysis

To prevent data leakage, the GDP series is shifted by one year before the rolling mean is computed. Without the shift, the average for year t would include the GDP of year t, which is the value being predicted. A five-year window is long enough to smooth a single exceptional year and short enough to retain 47 of the 52 years for modelling, since the first complete moving average is available for 1975. The rows for 1970–1974 are removed immediately before modelling, so that the exploratory analysis can use the full period.

4.7 Analytical Questions

Five analytical questions were investigated (Table 8). Question 2 is based on moving averages and Question 3 on lagged values.

Table 8. Analytical questions and techniques.

No.	Question	Technique
1	Does GDP show a consistent increasing trend over the decades?	Trend and growth rates
2	How does the five-year moving average smooth economic volatility?	Moving averages
3	Is the previous year's GDP a strong predictor of the current year's GDP?	Lagged values and autocorrelation
4	Which variables correlate most strongly with GDP growth?	Correlation of levels and of growth rates
5	How has income per person evolved relative to population growth?	Indexing to a common base year

Question 1: Does GDP show a consistent increasing trend over the decades?

This question examines whether the series is stable or trending and whether the trend is interrupted, since a trending series requires a model that can predict values above those observed during training.

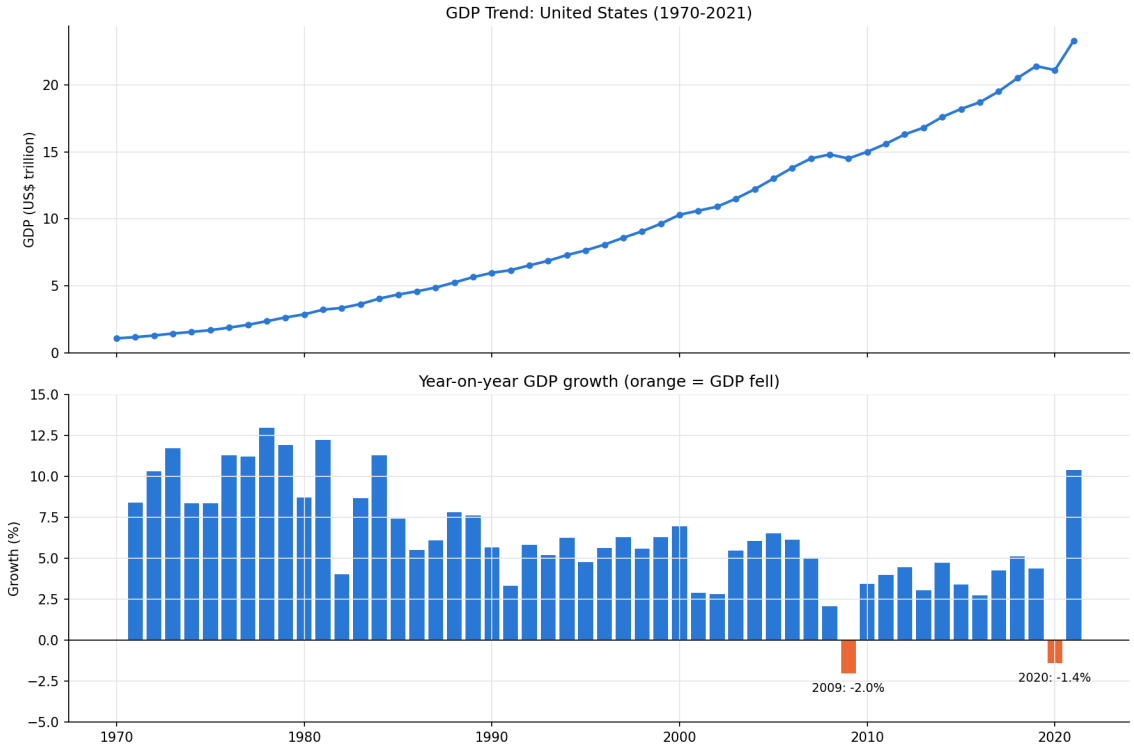


Figure 5. Top: US GDP, 1970–2021. Bottom: year-on-year growth; the two years in which GDP declined are marked.

GDP rose from US$1.07 trillion in 1970 to US$23.3 trillion in 2021, an increase by a factor of about 22, or 6.2% per year on average. The increase was steady: GDP grew in 49 of the 51 annual changes. It declined only in 2009 (−2.0%), following the 2008 financial crisis, and in 2020 (−1.4%), during the COVID-19 pandemic, and both declines were recovered within one to two years. The growth rate itself was not constant: nominal growth averaged 10.5% per year in 1971–1981, a period of high inflation, and about 4% after 2000. Because the test years (2012–2021) contain GDP values above any value in the training years, the model must be able to extrapolate.

Question 2: How does the five-year moving average smooth economic volatility?

This question examines how a rolling average removes year-to-year variation, and how far the moving-average feature lies from the value to be predicted.

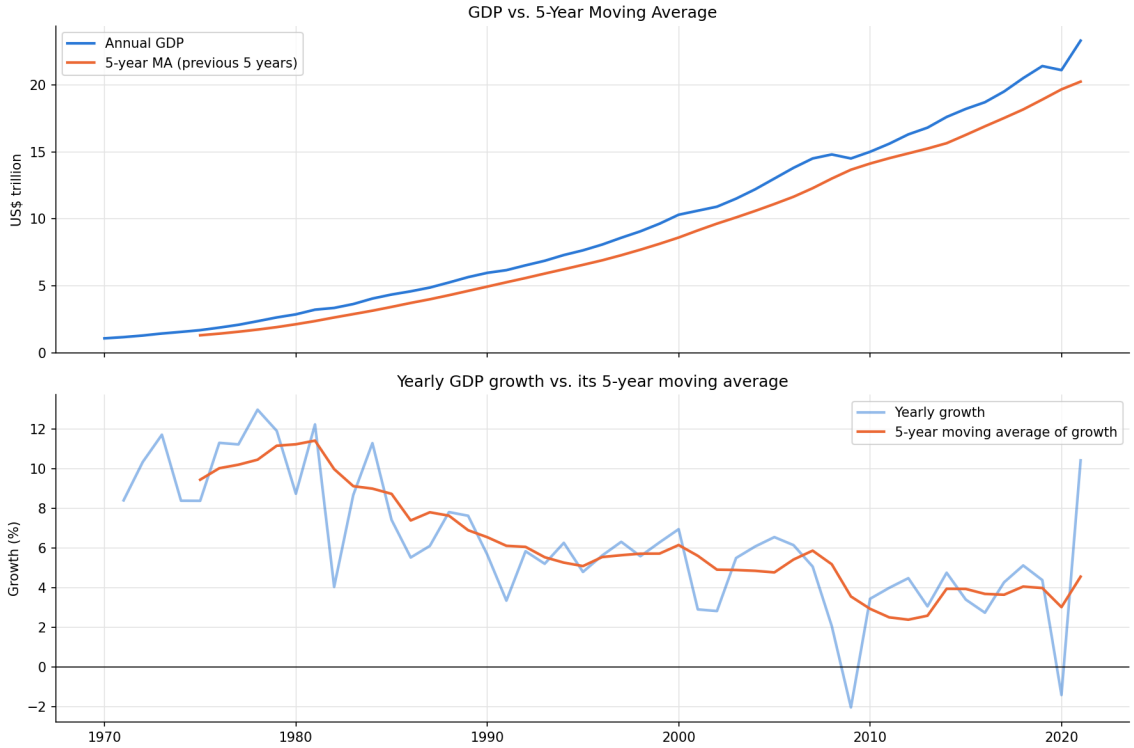


Figure 6. Top: annual GDP and the moving-average feature (mean of the five previous years). Bottom: annual growth and its five-year moving average.

On the level chart, the moving average forms a smooth line that continues to rise through the declines of 2009 and 2020. It lies below GDP by 19% on average, because on a rising trend the mean of the five previous years is always lower than the current value. The moving average alone would therefore under-predict, and the model must learn how far above the recent average GDP will lie. The growth chart shows the smoothing more clearly. Annual growth ranges from −2.0% to 13.0% (standard deviation 3.3 percentage points), whereas its five-year average ranges from 2.4% to 11.4% (standard deviation 2.5 points) and reveals long-run phases rather than individual shocks: about 10% in the 1970s, declining through the 1980s and 1990s to about 4% after 2000. The five-year moving average is therefore a stable summary of recent economic conditions, and it is used as a model feature alongside the previous year's GDP.

Question 3: Is the previous year's GDP a strong predictor of the current year's GDP?

This question tests for autocorrelation, the correlation of a series with a lagged copy of itself, for lags of one to ten years. Both the GDP level and the annual growth rate are examined, because a shared trend alone can make levels appear correlated.

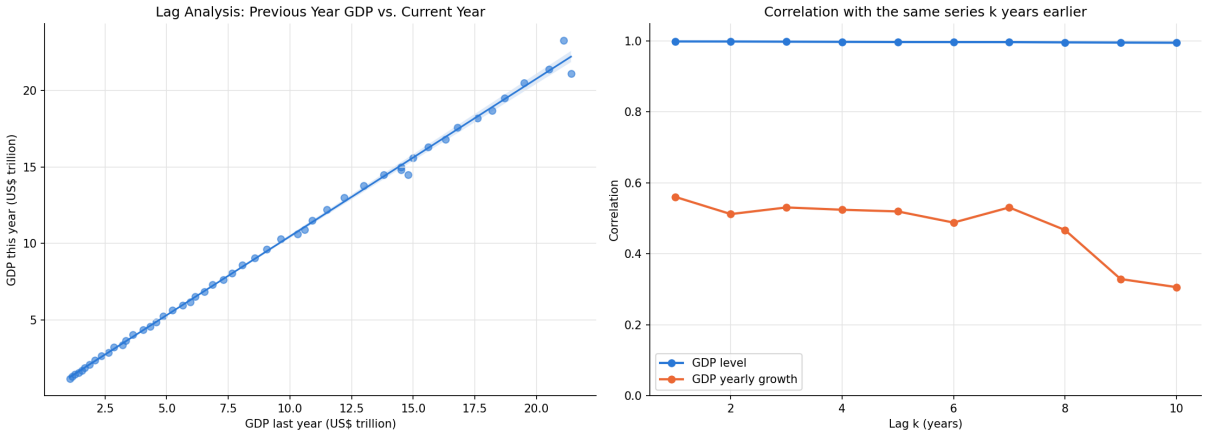


Figure 7. Left: GDP in the current year against GDP in the previous year. Right: correlation of GDP (level and growth rate) with the same series k years earlier.

Table 9. Autocorrelation of GDP at selected lags.

Lag (years)	**1**	**2**	**3**	**5**	**8**	**10**
GDP level	0.999	0.999	0.998	0.997	0.996	0.995
GDP growth rate	0.561	0.512	0.531	0.519	0.467	0.306

In levels, the correlation between the current and the previous year's GDP is 0.999, and the points lie almost exactly on a straight line. However, the level correlation remains above 0.995 even at a lag of ten years, which shows that much of it reflects the common trend rather than genuine predictive information. Growth rates remove the trend and give a more informative result: 0.56 at lag 1, about 0.5 up to lag 8 and 0.31 at lag 10. Strong years tend to be followed by strong years, partly because of the long phases identified in Question 2. GDP_Lag1 is therefore the most important feature, and it also defines the benchmark that any model must outperform: the naive forecast in which GDP in the current year equals GDP in the previous year.

Question 4: Which variables correlate most strongly with GDP growth?

This question identifies the explanatory variables that move with GDP. Because all of these series trend upwards, the correlations of their levels are compared with the correlations of their annual growth rates. A variable that correlates only in levels shares the trend but does not move with GDP from year to year.

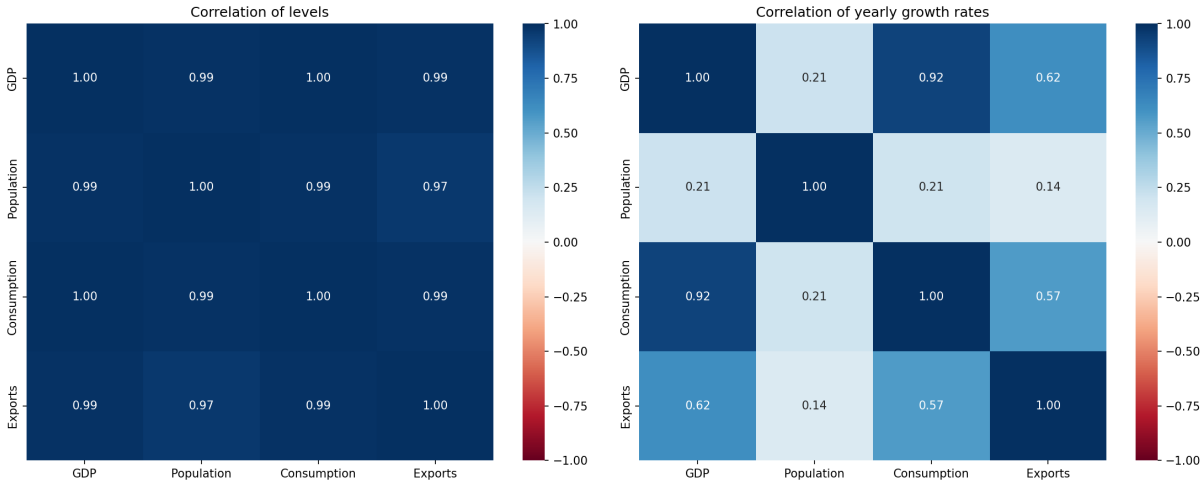


Figure 8. Correlation matrices of GDP, population, final consumption expenditure and exports. Left: levels. Right: annual growth rates.

Table 10. Correlation with GDP in levels and in growth rates.

Variable	**Levels**	**Growth rates**
Final consumption expenditure	1.00	0.92
Exports of goods and services	0.99	0.62
Population	0.99	0.21

In levels, every variable correlates with GDP at 0.99 or above, which would suggest that population is almost as informative as consumption. Growth rates distinguish between them. Consumption (0.92) grows faster when GDP grows faster, as expected for a component that accounts for about 80% of GDP. Exports (0.62) show a moderate relationship, consistent with trade being a smaller and more volatile part of GDP. Population (0.21) is only weakly related: it increases by a similar amount each year regardless of economic conditions, and its correlation of 0.99 in levels arises from the shared upward trend. Consumption is therefore retained as the strongest explanatory variable. Population is retained as a slow-moving measure of the size of the economy, although it is expected to contribute little.

Question 5: How has income per person evolved relative to population growth?

This question examines whether GDP grew mainly because the population increased or because income per person increased. GDP, GNI per capita and population are indexed to 1970 = 100 and plotted on a single logarithmic axis, so that the comparison does not depend on the scaling of separate axes.

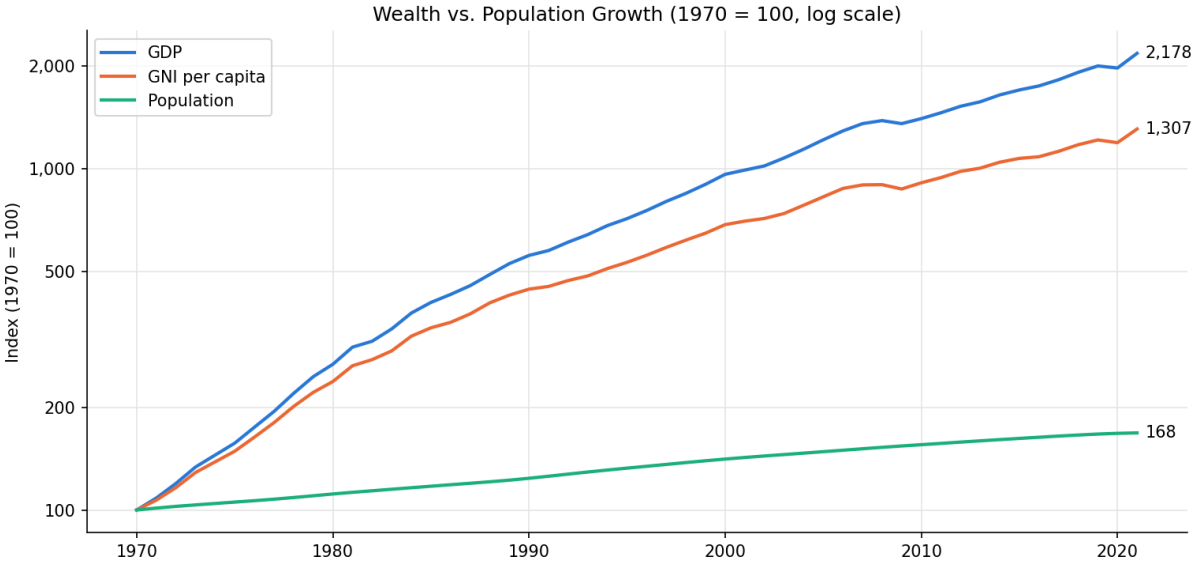


Figure 9. GDP, GNI per capita and population indexed to 1970 = 100 (logarithmic scale). The labels show the 2021 index values.

Between 1970 and 2021 the population grew by a factor of 1.68 (from 200 to 337 million), GNI per capita by a factor of 13.1 (from US$5,363 to US$70,081) and GDP by a factor of 21.8. The two components multiply: 1.68 × 13.1 ≈ 22. In logarithmic terms, population growth accounts for about 17% of GDP growth and higher income per person for the remaining 83%. Since the figures are in current dollars, part of the rise in income per person reflects inflation, but population growth remains far too slow to explain the growth in GDP. This agrees with the weak growth-rate correlation found in Question 4: year-to-year changes in GDP originate in income and spending, so recent GDP and consumption carry most of the predictive information.

4.8 Model Training and Experimental Design

The model uses four features selected from the analysis: GDP_Lag1 (Question 3), GDP_5Year_MA (Question 2), final consumption expenditure (Question 4) and population. The models and the tuning procedure were implemented with scikit-learn (Pedregosa et al., 2011). The experimental design is as follows:

1. The rows for 1970–1974, which lack a complete moving average, are removed, leaving 47 years.
2. The data are divided chronologically: 1975–2011 (37 years) for training and 2012–2021 (10 years) for testing. A random split would allow the model to be trained on later years and tested on earlier ones, which would leak future information.
3. Hyperparameters are tuned by grid search with TimeSeriesSplit (three splits), in which each validation fold contains only years that follow its training years. Candidate settings are ranked by root mean squared error (RMSE).
4. The naive forecast, in which GDP for year t equals GDP for year $t-1$, serves as the baseline (Hyndman & Athanasopoulos, 2021).
5. Performance is measured by RMSE, which expresses the typical error in US dollars and penalises large errors; by the mean absolute percentage error (MAPE); and by the coefficient of

determination (R^2).

Table 11. Experiments and hyperparameter search.

No.	Model	Rationale	Hyperparameters searched	Selected setting
0	Naive baseline	Benchmark from Question 3	None	None
1	Linear Regression	Able to extrapolate a trend (Question 1)	None (closed-form solution)	None
2	Decision Tree	Tests non-linear splitting rules	max_depth: 3, 5, 10, None; min_samples_split: 2, 5, 10	max_depth 5, min_samples_split 2
3	Random Forest	Averages many trees to reduce variance	n_estimators: 50, 100; max_depth: None, 5, 10	50 trees, max_depth None

Table 12. Results on the test years (2012–2021).

Model	RMSE (US$ billion)	MAPE (%)	R^2
Naive baseline	968	4.18	0.79
Linear Regression	889	4.17	0.82
Decision Tree (tuned)	4,298	18.38	−3.12
Random Forest (tuned)	4,427	19.15	−3.37

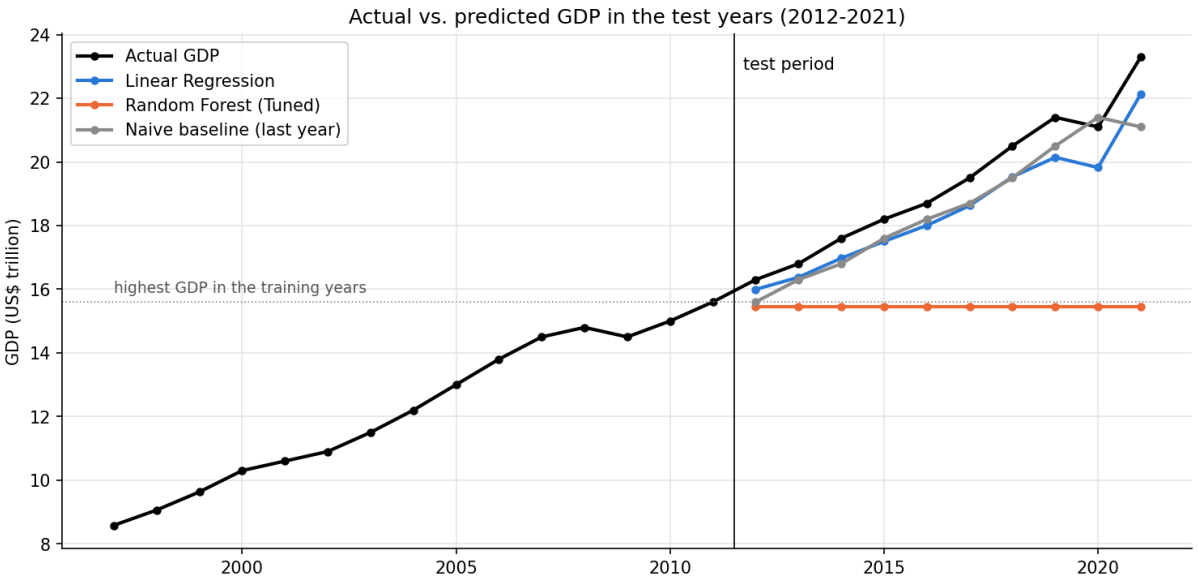


Figure 10. Actual GDP, 1997–2021, and the predictions of each model for the test years. The dotted line marks the highest GDP in the training years (2011).

Table 13. Prediction error by test year (% of actual GDP).

Model	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Linear Regression	−1.9	−2.5	−3.6	−3.8	−3.7	−4.5	−4.7	−5.9	−6.1	−5.0
Naive baseline	−4.3	−3.0	−4.5	−3.3	−2.7	−4.1	−4.9	−4.2	+1.4	−9.4

Linear Regression is the best-performing model and the only one that outperforms the baseline, with an RMSE 8% lower (US$889 billion compared with US$968 billion) and an R^2 of 0.82 compared with 0.79. Because it can extrapolate, it follows GDP above the range observed in training. The margin is modest, and the MAPE of the two is almost identical (about 4.2%), which confirms the finding of Question 3 that the previous year's GDP already contains most of the relevant information.

Both tree-based models perform poorly, and hyperparameter tuning does not correct this. A decision tree can only output averages of training values, and every test year has a higher GDP than the highest training year (US$15.6 trillion in 2011). Their predictions therefore remain flat just below that level while GDP continues to rise (Figure 10), which results in negative R^2 values, that is, a performance worse than predicting the mean of the test period.

Linear Regression under-predicts every test year, and the error widens from −1.9% in 2012 to −6.1% in 2020. This indicates that the relationship estimated from 1975–2011 becomes less accurate the further the model predicts beyond its training period. During the 2020–21 shock the model remains within about 6% of the actual value, because the same-year consumption figure falls and recovers with GDP. The naive baseline is 3–5% too low in ordinary years, over-predicts 2020 (+1.4%) and records the largest single error in 2021 (−9.4%), when GDP rebounded by 10%. Linear Regression obtains its lower RMSE mainly by avoiding this large error. Retraining the model each year as new data become available (walk-forward validation) would limit how far it extrapolates and is expected to reduce the growing bias.

Consumption and population enter the model with their values for the same year as the GDP being estimated, which is why the problem is defined as a nowcast. A strict one-year-ahead forecast would use the previous year's values of these indicators instead. Since consumption accounts for about 80% of GDP, such a variant would be expected to be less accurate and the naive baseline more difficult to outperform.

4.9 Summary of the Preprocessing Pipeline

Table 14 summarises the preprocessing steps. The same steps, with the exception of the exploratory analysis in step 5, are reused by the prediction script in Task 4 (Section 7).

Table 14. Preprocessing steps and their rationale.

Step	Operation	Rationale
1	Remove spaces from column names and text values	Padded names cause filters and look-ups to fail
2	Select the United States, sort by year, use Year as index	Largest complete economy without exchange-rate effects; time order is required for lags
3	Confirm one record per year without gaps or duplicates	Lag features assume that one row corresponds to one year
4	Linear interpolation inside the series, backward or forward fill at the edges	Preserves the trend; mean or zero imputation would create artificial spikes
5	Summary statistics, histograms and the ADF test	Revealed a strong trend, which guided the choice of features and models

Step	Operation	Rationale
6	GDP_Lag1: GDP of the previous year	Strongest single predictor (Question 3)
7	GDP_5Year_MA: mean of the five previous years, shifted before rolling	Smooths shocks (Question 2) without leaking the current year's GDP
8	Remove 1970–1974	Rows without a full five-year history cannot be used for training
9	Train on 1975–2011, test on 2012–2021, tune on time-ordered folds	The model must never learn from the future

5. Database Design and Implementation

This section describes Task 2. The records are stored in two databases: a relational database in MySQL and a document database in MongoDB. Both are created and loaded from the CSV file by Python scripts (task2/mysql_setup.py and task2/mongodb_setup.py), which also execute five queries each.

5.1 Relational Schema in MySQL

The relational design separates the data into three tables (Table 15). Country attributes that do not change over time, the name and the currency, are stored once in countries rather than repeated in each of the 10,512 yearly rows, which removes redundancy and keeps names consistent. Exchange rates are held in a separate table, exchange_rates, because they are monetary measures that describe the currency rather than the output of the economy, and most analytical queries do not require them. The yearly macroeconomic indicators, including GDP, form the central time-series table, economic_indicators.

Table 15. Tables of the relational schema.

Table	Primary key	Foreign key	Content	Rows
countries	country_id	None	Name and currency of each country	220
exchange_rates	id	country_id references countries	AMA and IMF-based exchange rates per country and year	10,512
economic_indicators	id	country_id references countries	20 yearly indicators, including GDP, population and consumption	10,512

Each relationship is one-to-many: one country has many yearly exchange-rate records and many yearly indicator records (Figure 11). A unique constraint on (country_id, year) in both dependent tables guarantees one record per country and year, a property on which the time-series features rely. Indexes on year and on (country_id, year) support the most frequent access patterns of the API: retrieving the records of one country in year order, the latest year and a range of years. Listing 1 shows the data definition statements.

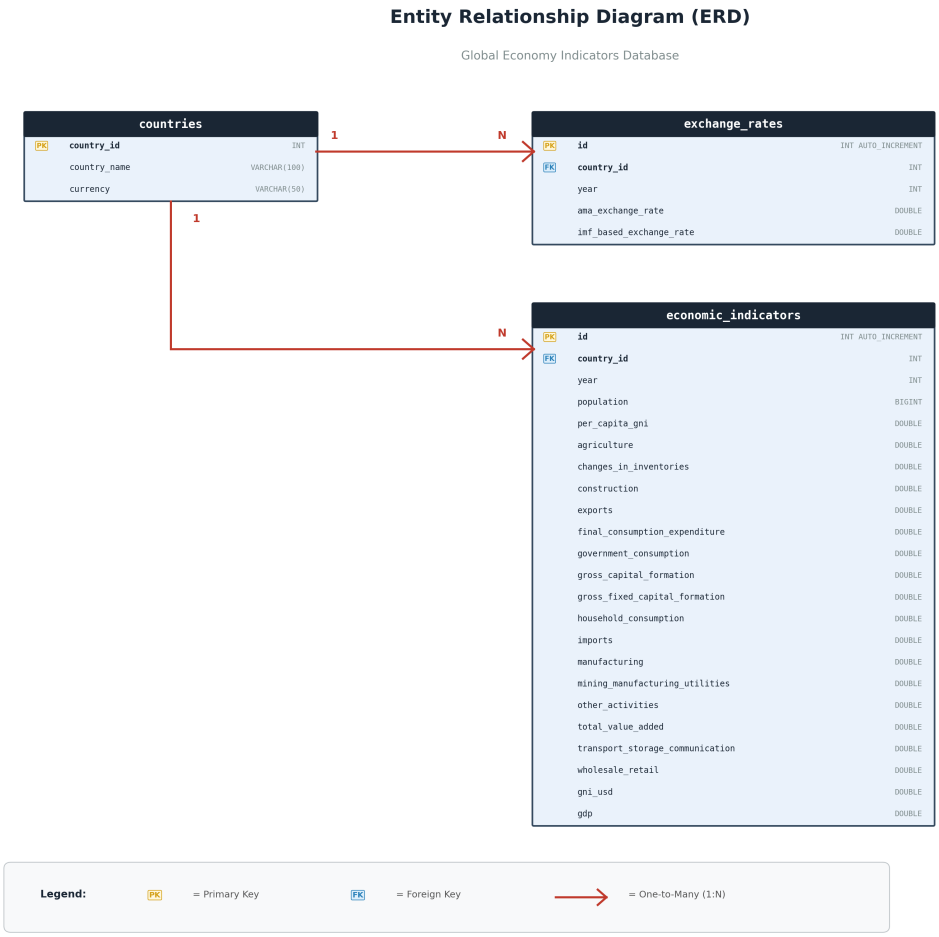


Figure 11. Entity relationship diagram of the MySQL database. Each country has many exchange-rate records and many economic-indicator records.

Listing 1. Data definition statements of the relational schema (abridged; full version in task2/sql_schema.sql).

```
CREATE TABLE countries (
    country_id   INT PRIMARY KEY,
    country_name VARCHAR(100) NOT NULL,
    currency     VARCHAR(50)
);

CREATE TABLE exchange_rates (
    id INT AUTO_INCREMENT PRIMARY KEY,
    country_id INT NOT NULL,
    year INT NOT NULL,
    ama_exchange_rate DOUBLE,
    imf_based_exchange_rate DOUBLE,
    FOREIGN KEY (country_id) REFERENCES countries(country_id),
    UNIQUE KEY unique_country_year (country_id, year)
);

CREATE TABLE economic_indicators (
    id INT AUTO_INCREMENT PRIMARY KEY,
    country_id INT NOT NULL,
    year INT NOT NULL,
    population BIGINT,
    final_consumption_expenditure DOUBLE,
    gdp DOUBLE,
    -- 17 further indicator columns
    FOREIGN KEY (country_id) REFERENCES countries(country_id),
    UNIQUE KEY unique_country_year (country_id, year)
);

CREATE INDEX idx_economic_year ON economic_indicators(year);
CREATE INDEX idx_economic_country_year ON economic_indicators(country_id, year);
CREATE INDEX idx_exchange_year ON exchange_rates(year);
```

The loading script removes the padding spaces from the names and inserts the countries first, because the other two tables reference them, before inserting the exchange rates and the indicators. Missing values are stored as NULL rather than zero, so that they cannot be mistaken for genuine observations. The use of INSERT IGNORE together with the unique constraints prevents duplicate country-year rows if the script is run more than once.

Figures 12 and 13 show the output of the setup script when it is run against the MySQL server: the database and tables are created, 220 countries and 10,512 records are loaded, and the five queries described in Section 5.3 are executed.

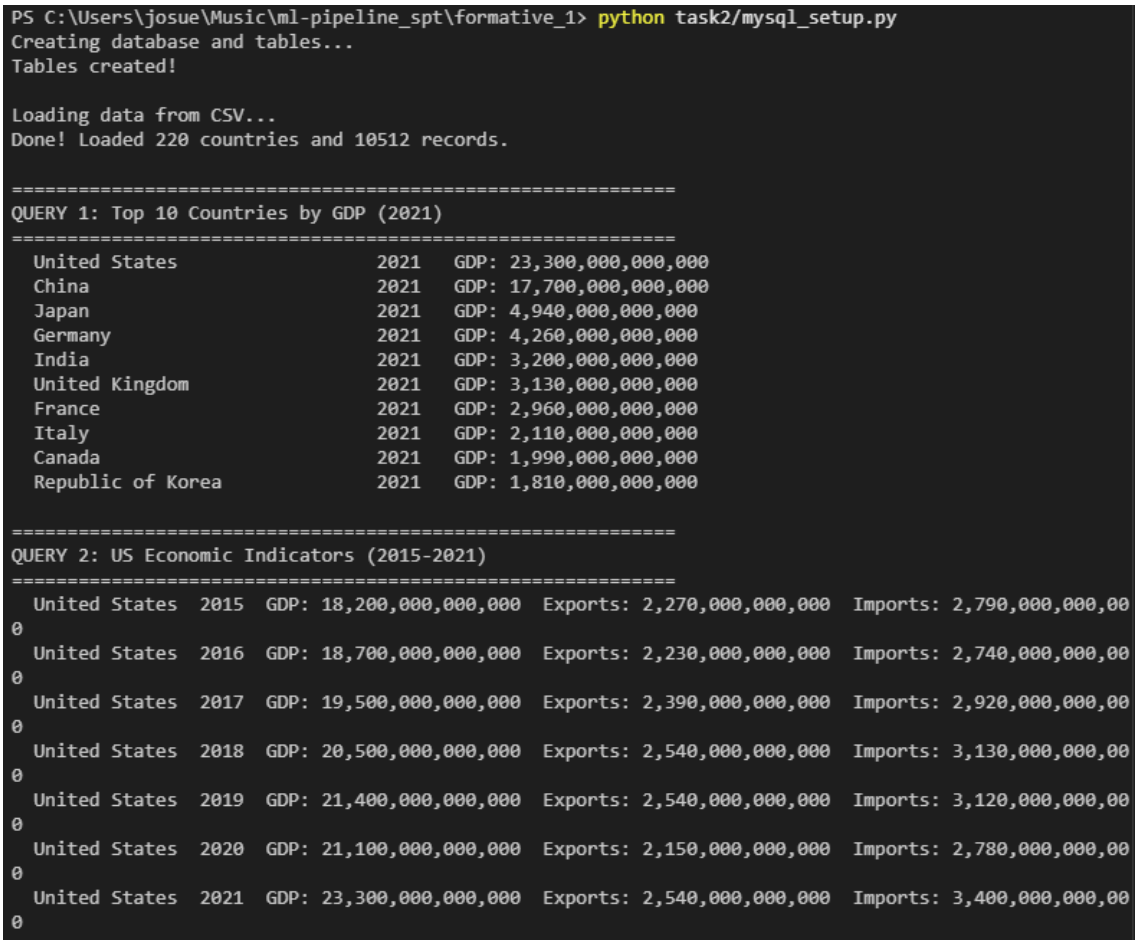


Figure 12. Output of python task2/mysql_setup.py: creating the tables, loading 220 countries and 10,512 records, and MySQL Queries 1 and 2.

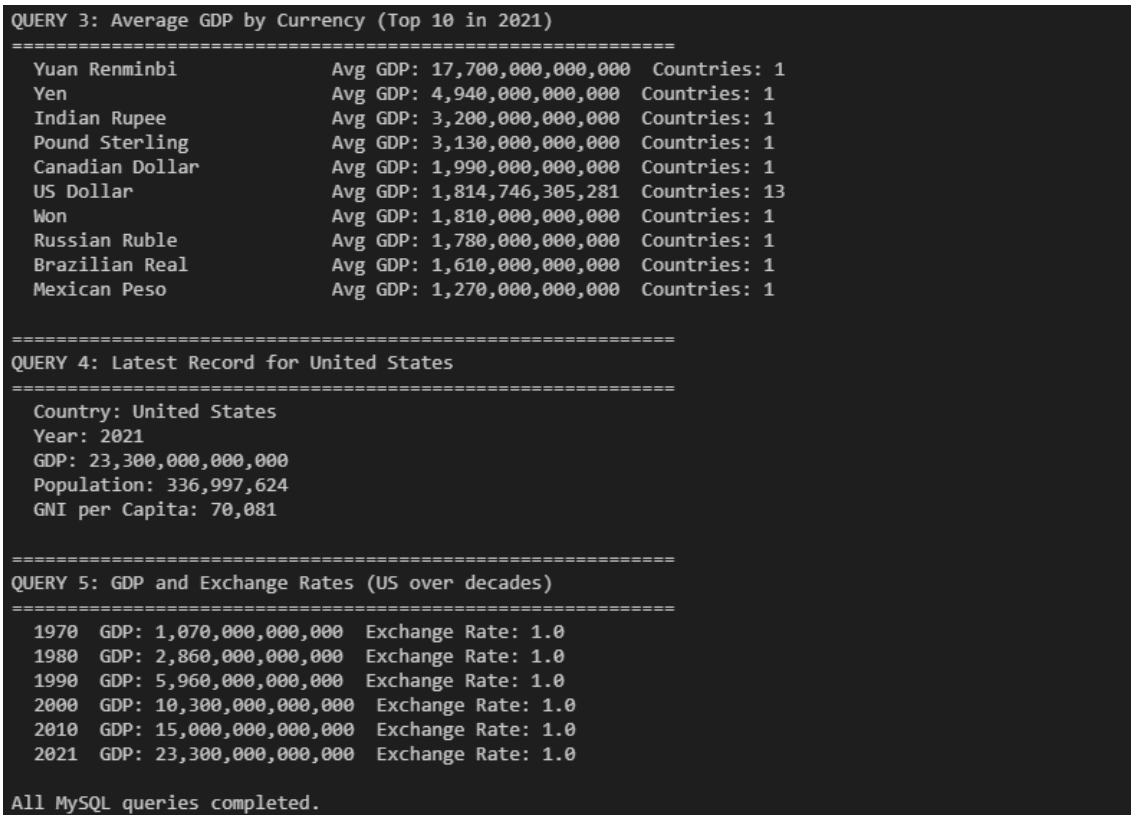


Figure 13. Output of MySQL Queries 3 to 5: the average GDP by currency in 2021, the latest US record and US GDP with exchange rates by decade.

5.2 Document Design in MongoDB

MongoDB stores the same data in a single collection, economic_records, with one document per country and year (10,512 documents). Instead of distributing the data across tables, each document groups related fields into four embedded sub-documents: exchange_rates, demographics, economic_indicators and sector_breakdown. This design reflects the way in which the data are read: the API and the prediction script always retrieve complete country-year records, which in MongoDB requires a single document read and no joins. The country name and currency are repeated in each document; document databases accept this redundancy in exchange for simpler and faster reads. A compound index on (country_name, year) and an index on year support the same access patterns as the MySQL indexes. Listing 2 shows the document for the United States in 2021, as stored in the database.

Listing 2. Document for the United States in 2021 (the _id field is omitted).


```
{
  "country_id": 840,
  "country_name": "United States",
  "currency": "US Dollar",
  "year": 2021,
  "exchange_rates": {
    "ama_exchange_rate": 1.0,
    "imf_based_exchange_rate": 1.0
  },
  "demographics": {
    "population": 336997624,
    "per_capita_gni": 70081.0
  },
  "economic_indicators": {
    "gdp": 23300000000000.0,
    "gni_usd": 23600000000000.0,
    "exports": 2540000000000.0,
    "imports": 3400000000000.0,
    "final_consumption_expenditure": 19300000000000.0,
    "government_consumption": 3350000000000.0,
    "household_consumption": 15900000000000.0,
    "gross_capital_formation": 4920000000000.0,
    "gross_fixed_capital_formation": 4940000000000.0,
    "changes_in_inventories": -19091000000.0
  },
  "sector_breakdown": {
    "agriculture": 207000000000.0,
    "manufacturing": 2500000000000.0,
    "mining_manufacturing_utilities": 3270000000000.0,
    "construction": 945000000000.0,
    "transport_storage_communication": 2490000000000.0,
    "wholesale_retail": 3520000000000.0,
    "other_activities": 12900000000000.0,
    "total_value_added": 23300000000000.0
  }
}
```


Figure 14 shows the output of the setup script: the 10,512 documents are inserted, the indexes are created and the first sample document (Afghanistan, 1970) is printed.

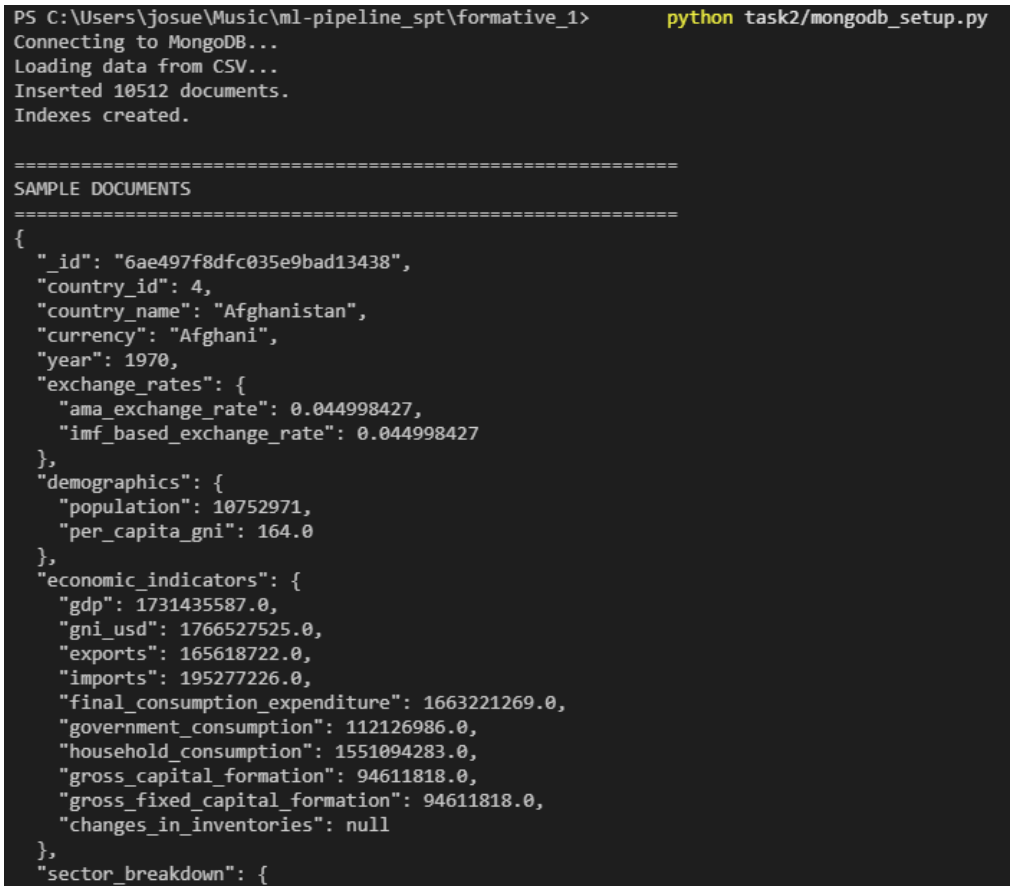


Figure 14. Output of python task2/mongodb_setup.py: loading the collection, creating the indexes and printing the first sample document.

5.3 Queries

Five queries were implemented in each database (Table 16). They cover ranking, filtering by a date range, aggregation, retrieval of the latest record and the combination of data from several sources.

Table 16. Queries implemented in each database.

No.	MySQL	MongoDB
1	Ten largest economies in 2021 (JOIN, ORDER BY, LIMIT)	Ten largest economies in 2021 (find, sort, limit)
2	US indicators for 2015–2021 (WHERE ... BETWEEN)	US records for 2015–2021 (find with $gte and $lte)
3	Average GDP by currency in 2021 (GROUP BY, AVG)	Number of records per country ($group)
4	Latest US record (ORDER BY year DESC LIMIT 1)	Latest US record (find_one sorted by year, descending)
5	US GDP and exchange rates by decade (join of three tables)	Average GNI per capita by currency in 2021 ($match, $group)

Query 2 is the time-series query on which the date-range endpoint of the API is based. In MySQL it joins the indicators to the country names and filters the years with BETWEEN (Listing 3); in

MongoDB the same filter is expressed with the $gte and $lte operators on the indexed year field (Listing 4). When executed, Query 1 returns the United States (US$23.3 trillion) and China (US$17.7 trillion) as the two largest economies in 2021, and Query 4 returns the 2021 record shown in Listing 2.

Listing 3. Query 2 in MySQL: US indicators for a range of years.

```
SELECT c.country_name, e.year, e.gdp, e.exports, e.imports
FROM economic_indicators e
JOIN countries c ON e.country_id = c.country_id
WHERE c.country_name = 'United States'
  AND e.year BETWEEN 2015 AND 2021
ORDER BY e.year;
```

Listing 4. Query 2 in MongoDB (Python, PyMongo).

```
collection.find(
    {"country_name": "United States", "year": {"$gte": 2015, "$lte": 2021}},
    {"_id": 0, "year": 1, "economic_indicators.gdp": 1,
     "economic_indicators.exports": 1, "demographics.population": 1},
).sort("year", 1)
```

Figures 15 and 16 show the output of the five MongoDB queries when the setup script is run.

```
============================================================
QUERY 1: Top 10 Countries by GDP (2021)
============================================================
  United States                 GDP: 23,300,000,000,000
  China                         GDP: 17,700,000,000,000
  Japan                         GDP: 4,940,000,000,000
  Germany                       GDP: 4,260,000,000,000
  India                         GDP: 3,200,000,000,000
  United Kingdom                GDP: 3,130,000,000,000
  France                        GDP: 2,960,000,000,000
  Italy                         GDP: 2,110,000,000,000
  Canada                        GDP: 1,990,000,000,000
  Republic of Korea             GDP: 1,810,000,000,000

============================================================
QUERY 2: US Records by Date Range (2015-2021)
============================================================
  2015  GDP: 18,200,000,000,000  Exports: 2,270,000,000,000  Population: 324,607,776
  2016  GDP: 18,700,000,000,000  Exports: 2,230,000,000,000  Population: 327,210,198
  2017  GDP: 19,500,000,000,000  Exports: 2,390,000,000,000  Population: 329,791,231
  2018  GDP: 20,500,000,000,000  Exports: 2,540,000,000,000  Population: 332,140,037
  2019  GDP: 21,400,000,000,000  Exports: 2,540,000,000,000  Population: 334,319,671
  2020  GDP: 21,100,000,000,000  Exports: 2,150,000,000,000  Population: 335,942,003
  2021  GDP: 23,300,000,000,000  Exports: 2,540,000,000,000  Population: 336,997,624

============================================================
QUERY 3: Records per Country (Top 10)
============================================================
  Gambia                        Records: 52
  Tunisia                       Records: 52
  Bermuda                       Records: 52
  New Caledonia                 Records: 52
  Madagascar                    Records: 52
  Denmark                       Records: 52
  Guinea                        Records: 52
  Benin                         Records: 52
  Norway                        Records: 52
  Antigua and Barbuda           Records: 52
```

Figure 15. Output of MongoDB Queries 1 to 3: the ten largest economies in 2021, the US records for 2015–2021 and the number of records per country.



Figure 16. Output of MongoDB Queries 4 and 5: the latest US record and the average GNI per capita by currency in 2021.

5.4 Comparison of the Two Designs

Table 17 compares the two designs. The relational schema is better suited to enforcing consistency and to queries that combine data across tables, whereas the document model is better suited to reading complete records quickly.

Table 17. Comparison of the MySQL and MongoDB designs.

Aspect	**MySQL**	**MongoDB**
Structure	Three normalised tables linked by foreign keys	One collection of documents with embedded sub-documents
Reading one country-year record	Join of the three tables	Single document read
Data integrity	Enforced by primary keys, foreign keys and unique constraints	Enforced by the application
Changes to the structure	Require ALTER TABLE statements	New fields can be added to new documents
Most suitable for	Consistent related data and complex joins	Fast retrieval of complete records

6. API for CRUD Operations and Time-Series Queries

This section describes Task 3. The API is implemented with FastAPI in task3/api_demo.py. It exposes the same endpoints for both databases, under the prefixes /api/sql and /api/mongodb, so that a client can change databases by changing only the prefix. Request bodies are validated with Pydantic models: EconomicRecordCreate requires the country name and year and accepts all indicators as optional fields, while EconomicRecordUpdate contains only optional fields, which

allows partial updates. FastAPI also generates interactive documentation at /docs. Table 18 lists the endpoints.

Table 18. Endpoints of the API (identical for /api/sql and /api/mongodb).

Method	Path	Operation	MySQL implementation	MongoDB implementation
POST	/records	Create a record	Inserts into economic_indicators (and exchange_rates when rates are given); adds the country if it does not exist	insert_one with the nested structure of Listing 2
GET	/records	Read all records of a country (parameter: country)	SELECT with joins, ordered by year	find, sorted by year
GET	/records/latest	Read the latest record of a country (parameter: country)	ORDER BY year DESC LIMIT 1	find_one, sorted by year in descending order
GET	/records/range	Read records in a date range (parameters: country, start_date, end_date)	year BETWEEN the start and end years	year with $gte and $lte
PUT	/records/{id}	Update a record	UPDATE of the supplied fields only, in the table that holds each field	update_one with $set on the nested paths
DELETE	/records/{id}	Delete a record	Deletes the indicator row and the matching exchange-rate row	delete_one by document identifier

Several design decisions apply to both databases. All read endpoints return the same flat JSON structure, including a Date field in YYYY-MM-DD format derived from the year, so the prediction script can process the output of either database without modification. Country names are matched without regard to letter case. In MySQL, inserts, updates and deletions that involve two tables are executed in one transaction and rolled back if any statement fails. Errors are reported with standard HTTP status codes: 400 for invalid input, such as a malformed date, an end date before the start date or an update without fields; 404 when no record matches; and 503 when the database cannot be reached.

Figure 17 shows the interactive documentation generated by FastAPI, which lists the endpoints and allows each of them to be tested from a web browser. Figures 18 and 19 show a request to the date-range endpoint of the MongoDB database, made through this documentation, and the response returned by the API.

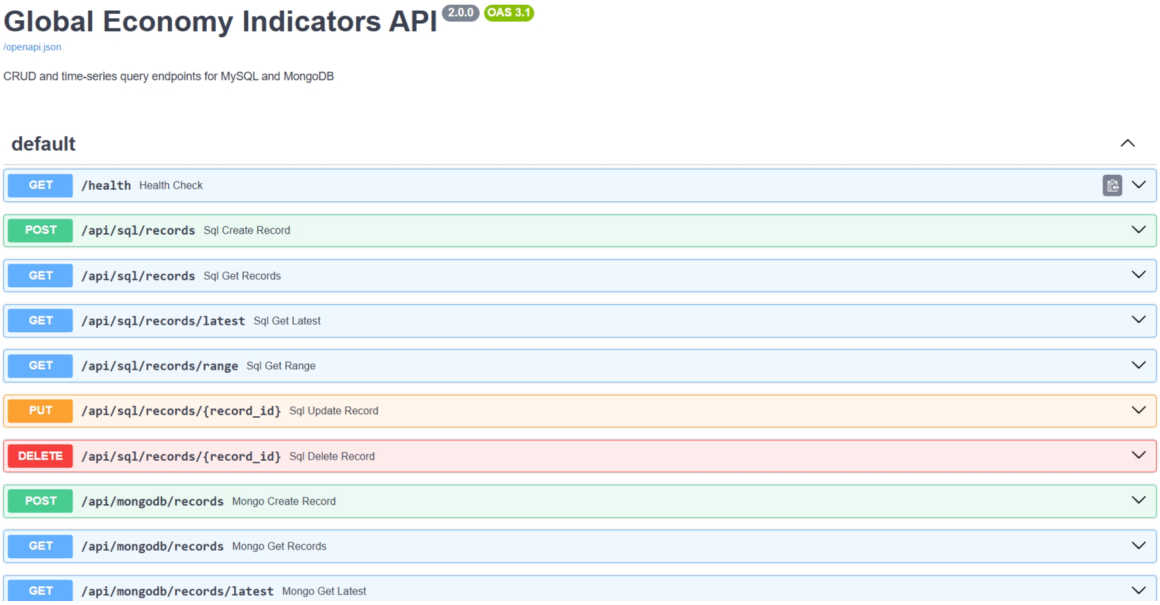


Figure 17. Interactive API documentation generated by FastAPI at /docs, showing the health check, the six MySQL endpoints and the first MongoDB endpoints; the remaining MongoDB endpoints follow on the same page.

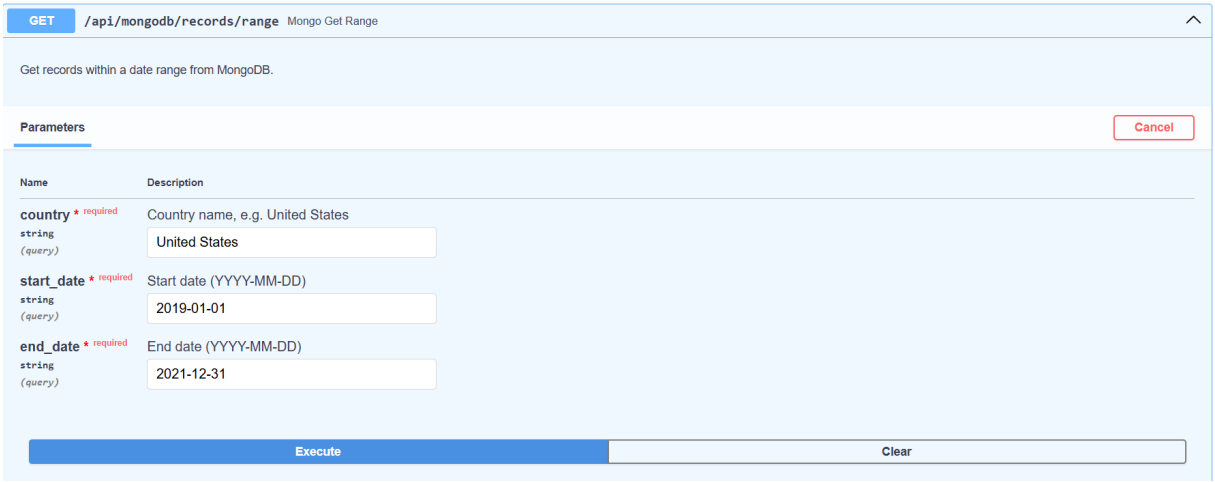


Figure 18. Request to GET /api/mongodb/records/range for the United States from 2019-01-01 to 2021-12-31, entered in the interactive documentation.

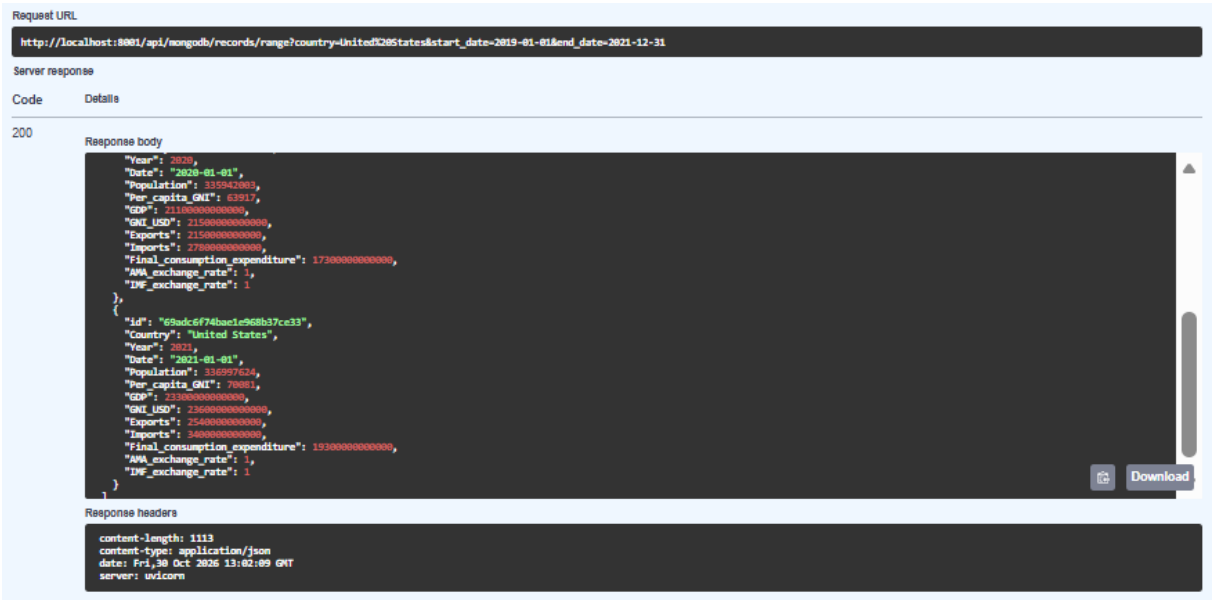


Figure 19. Response to the request in Figure 18: status code 200 and a response body with one record per year (the records for 2020 and 2021 are visible).

Listings 5 and 6 show further responses obtained from the running API connected to the MongoDB database.

Listing 5. Request for the latest US record and the response.

```
GET /api/mongodb/records/latest?country=United%20States

HTTP 200
{
  "country": "United States",
  "data": {
    "id": "69adc6f74bae1e968b37ce33",
    "Country": "United States",
    "Year": 2021,
    "Date": "2021-01-01",
    "Population": 336997624,
    "Per_capita_GNI": 70081.0,
    "GDP": 23300000000000.0,
    "GNI_USD": 23600000000000.0,
    "Exports": 2540000000000.0,
    "Imports": 3400000000000.0,
    "Final_consumption_expenditure": 19300000000000.0,
    "AMA_exchange_rate": 1.0,
    "IMF_exchange_rate": 1.0
  }
}
```

Listing 6. Error responses for an invalid date range and an unknown country.

```
GET /api/mongodb/records/range?country=United%20States
    &start_date=2021-01-01&end_date=2015-12-31
HTTP 400   {"detail": "end_date must be after start_date"}

GET /api/mongodb/records/latest?country=Atlantis
HTTP 404   {"detail": "No records found for country: Atlantis"}
```

7. Prediction Script

This section describes Task 4. The prediction script, task4/predict_from_api.py, connects the components described above and performs the four steps listed in Table 19.

Table 19. Steps of the prediction script.

Step	Operation	Description
1	Fetch	Requests the records of a country from the API: all records, or a date range through /records/range when start and end dates are given. MongoDB is used by default and MySQL with the option --db sql
2	Preprocess	Converts the JSON response into a table indexed by year and applies the Task 1 preprocessing: one row per year, interpolation of missing values, GDP_Lag1 and GDP_5Year_MA
3	Load the model	Loads the Linear Regression model and the feature order saved by task4/train_model.py, and checks that both come from the same training run
4	Predict	Estimates GDP for the latest year, or for the year given with --year, and compares the estimate with the actual value, where available, and with the naive baseline

7.1 Consistency with Task 1

To ensure that the model receives exactly the same inputs as in Task 1, the cleaning and feature-engineering steps are implemented once, in task4/preprocessing.py, and used by both the training script and the prediction script. The training script, task4/train_model.py, trains the model on the same data, features and chronological split as the notebook and reproduces its test results (RMSE US$889 billion, MAPE 4.17%, R^2 0.82). It saves the model with joblib, together with the feature order, the preprocessing configuration and a summary of the training run. GDP is interpolated only between known years, so a year without a GDP value, for example a new record created through the API, remains empty and can be predicted.

7.2 Example Run

Listing 7 shows the output of the script when it requests the US records for 2010–2021 from the running API through the date-range endpoint.

Listing 7. Output of python task4/predict_from_api.py --start_date 2010-01-01 --end_date 2021-12-31.

```
[1/4] Fetching records for United States from http://localhost:8000/api/mongodb/records/range
      12 records received
[2/4] Preprocessing (same pipeline as Task 1: interpolation, GDP_Lag1, GDP_5Year_MA)
      Years 2010-2021, one row per year
[3/4] Loading the trained model
      Features: ['Population', 'GDP_Lag1', 'GDP_5Year_MA', 'Final_consumption_expenditure']
[4/4] Making the prediction

=== GDP estimate: United States, 2021 ===
Model: LinearRegression trained on 1975-2011 (test R2 0.824, MAPE 4.17%)
Inputs:
  Population                                        336,997,624
  GDP_Lag1 (GDP 2020)                          US$21.10 trillion
  GDP_5Year_MA (mean GDP 2016-2020)            US$20.24 trillion
  Final_consumption_expenditure                US$19.30 trillion

Predicted GDP 2021: US$22.14 trillion
Actual GDP 2021 (from API): US$23.30 trillion -> model error -5.0%
Naive baseline (GDP 2020): US$21.10 trillion -> baseline error -9.4%
```

The model estimates 2021 GDP at US$22.14 trillion against an actual value of US$23.30 trillion, an error of −5.0%, whereas the naive baseline has an error of −9.4%. Both values are identical to the 2021 column of Table 13, which confirms that the path from the API through preprocessing to the model reproduces the results of Task 1. Figure 20 shows a second run, in which the script fetched all 52 US records through the /records endpoint and produced the same estimate.

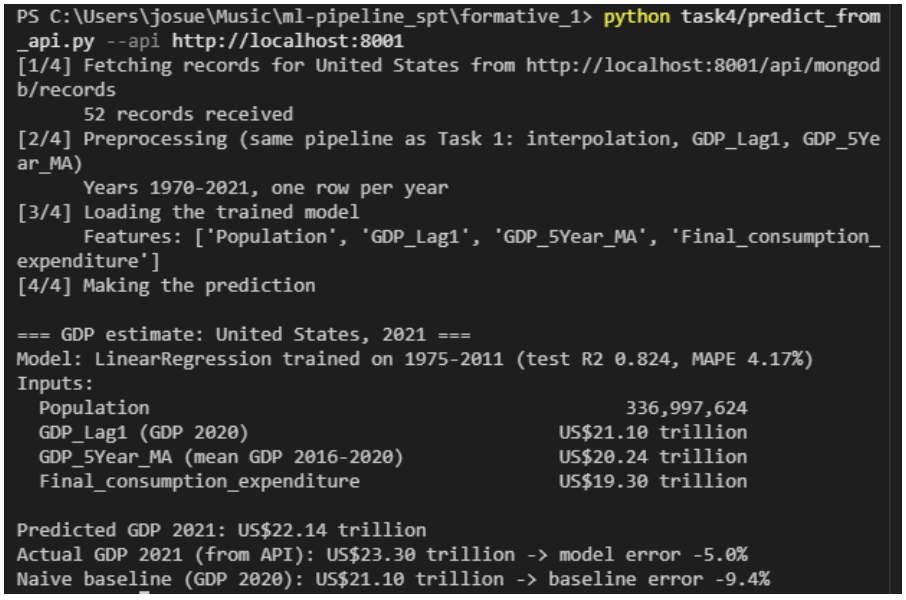


Figure 20. Terminal output of python task4/predict_from_api.py --api http://localhost:8001. The script fetched all 52 US records (1970–2021) and estimated 2021 GDP at US$22.14 trillion, an error of −5.0% against −9.4% for the naive baseline.

7.3 Error Handling

The script stops with an explanatory message and exit code 1 when the API is unavailable or returns an error (for example, for an unknown country), when only one of the two dates is supplied, when fewer than six consecutive years are available (the five-year moving average requires five previous years), when the requested year is not among the records, or when the model has not been trained. If

the population or consumption value for the year being estimated is missing, it is filled from neighbouring years, as in Task 1, and a warning is printed.

8. Conclusion

This project implemented a complete pipeline for annual time-series data, from exploratory analysis to a prediction service. The analysis showed that US GDP follows a strong non-stationary trend, that the previous year's GDP and same-year consumption carry most of the predictive information, and that population correlates with GDP mainly through the shared trend. Of the models tested, only Linear Regression outperformed the naive baseline (RMSE US$889 billion compared with US$968 billion), because it can extrapolate beyond the training range, whereas the tree-based models cannot. The data are stored in a normalised MySQL schema and in a MongoDB collection of embedded documents, served through a FastAPI interface with identical endpoints for both databases, and used by a prediction script that reproduces the results of Task 1 exactly.

The main limitations are the small number of annual observations, the use of nominal values, which include inflation, and the increasing under-prediction in the later test years. Future work could retrain the model each year with walk-forward validation, use inflation-adjusted GDP, extend the analysis to other countries and add authentication to the API before it is made available beyond a local environment.

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